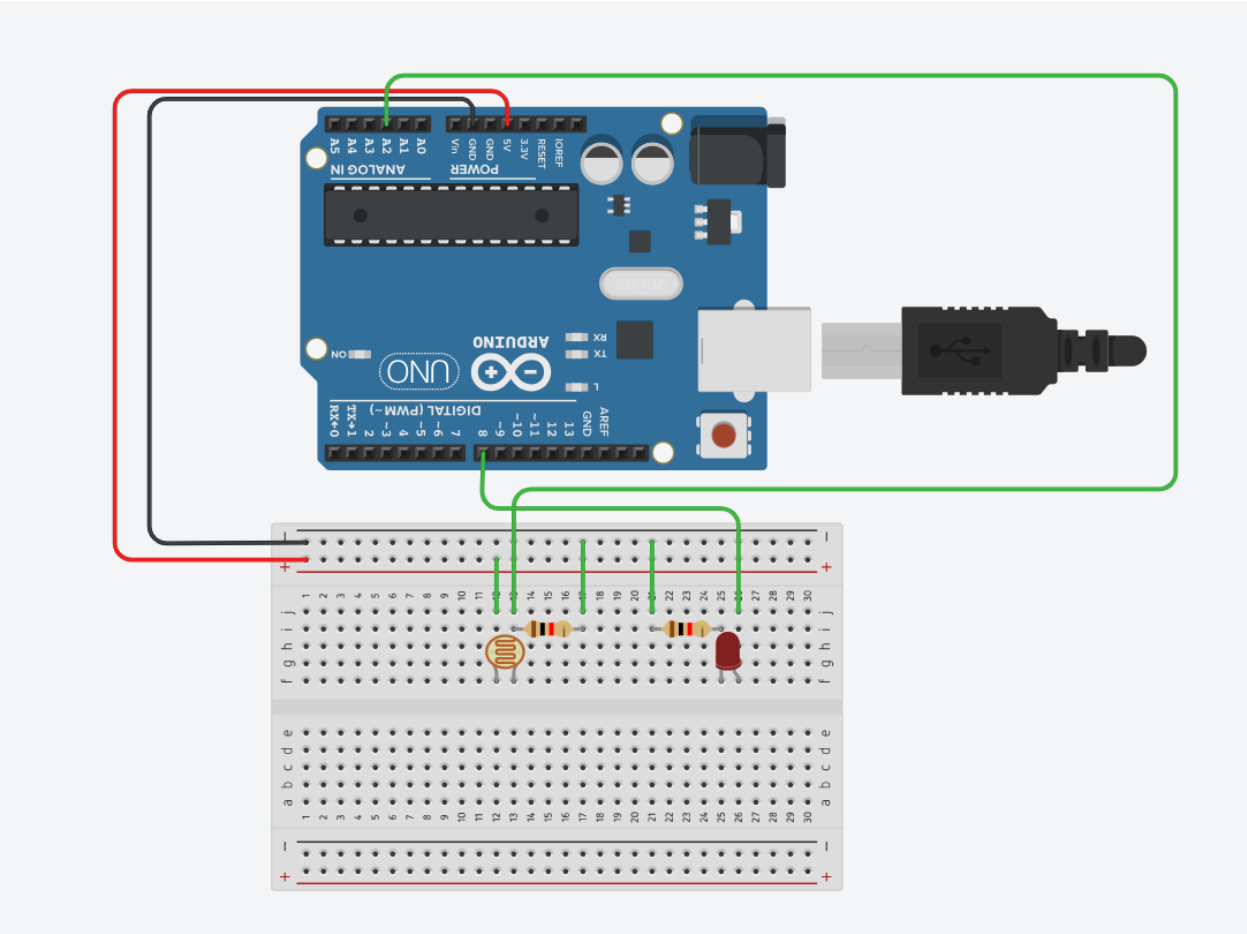
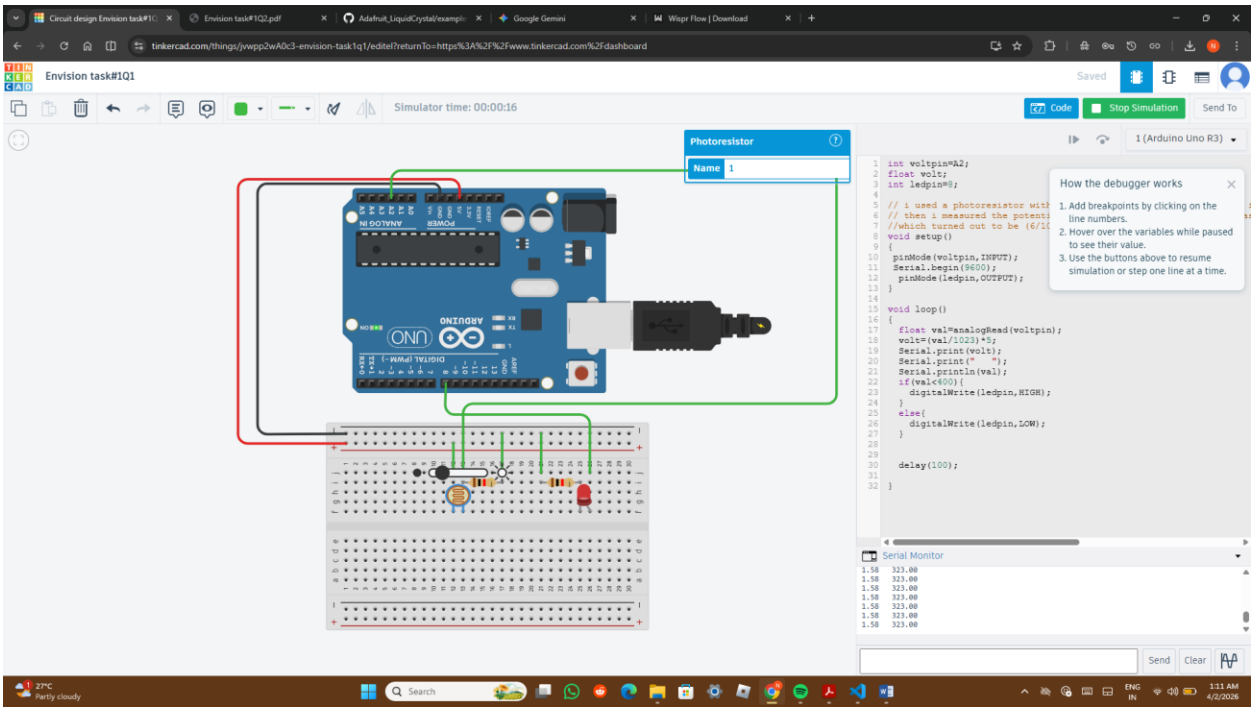
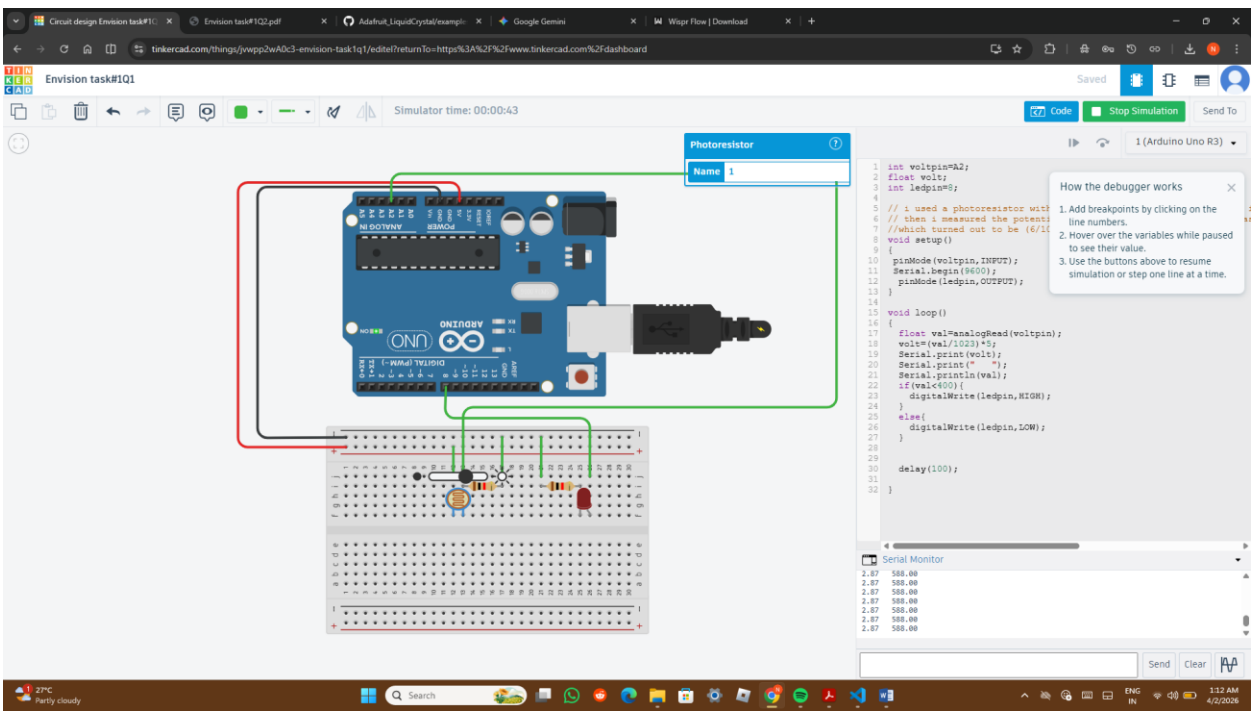
TASK 1 SOLUTIONS:

I used a photo resistor in series with a normal resistance and then ran a test ckt in which I measured the potential drop a/c the normal resistor as I went from min possible value of light to maximum possible value of light(the values were 6 and 679). Then I used basic if else statements to turn the led on after a certain threshold.

Ckt:







int voltpin=A2;

float volt;

int ledpin=8;

void setup()

```
{
 pinMode(voltpin,INPUT);
 Serial.begin(9600);
  pinMode(ledpin,OUTPUT);
}
void loop()
{
  float val=analogRead(voltpin);
  volt=(val/1023)*5;
  Serial.print(volt);
  Serial.print("   ");
  Serial.println(val);
  if(val<400){
    digitalWrite(ledpin,HIGH);
  }
  else{
    digitalWrite(ledpin,LOW);
  }
  delay(100);
}
```

https://www.tinkercad.com/things/jvwpp2wA0c3-envision-task1q1

TASK 2 SOLUTION:

I used the adafruit github repo for the functions.

```
#include <Adafruit_LiquidCrystal.h>
Adafruit_LiquidCrystal lcd(0);
int pin1=9;
```

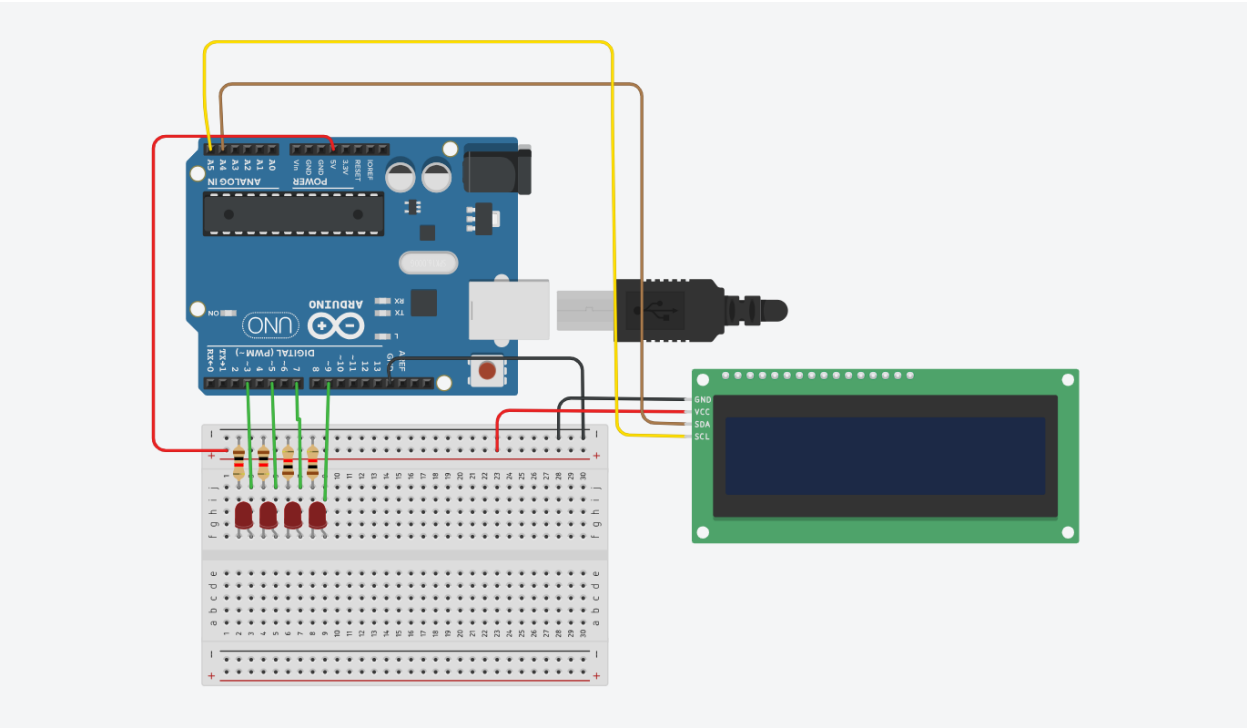
```
int pin2=7;
int pin3=5;
int pin4=3;
int num1,num2;

void setup() {
  Serial.begin(9600);
  lcd.begin(16, 2);
  lcd.setBacklight(HIGH);
  lcd.print("hello, world!");
  delay(2000);
  lcd.clear();

  pinMode(pin1, OUTPUT);
  pinMode(pin2, OUTPUT);
  pinMode(pin3, OUTPUT);
  pinMode(pin4, OUTPUT);




}

void loop() {
  Serial.println("Enter your number1: ");
  while(Serial.available()==0){
  }
  num1=Serial.parseInt();
  Serial.println(num1);
  Serial.println("Enter your number2: ");
```

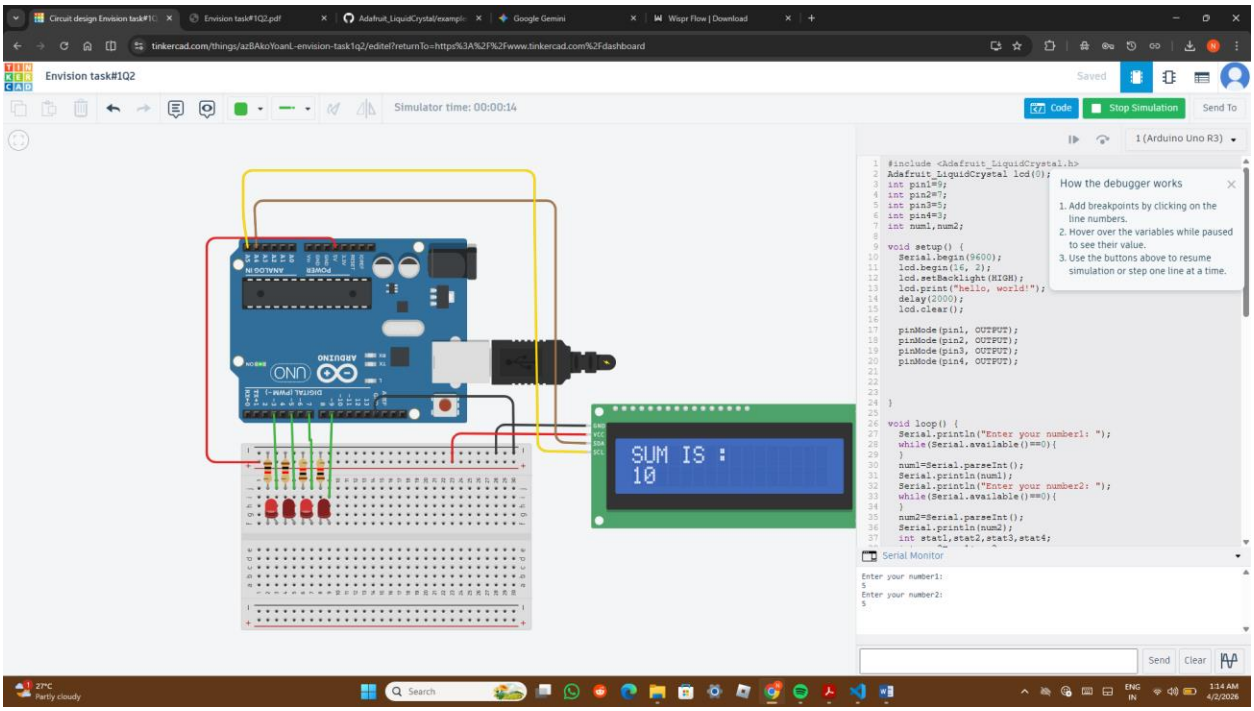
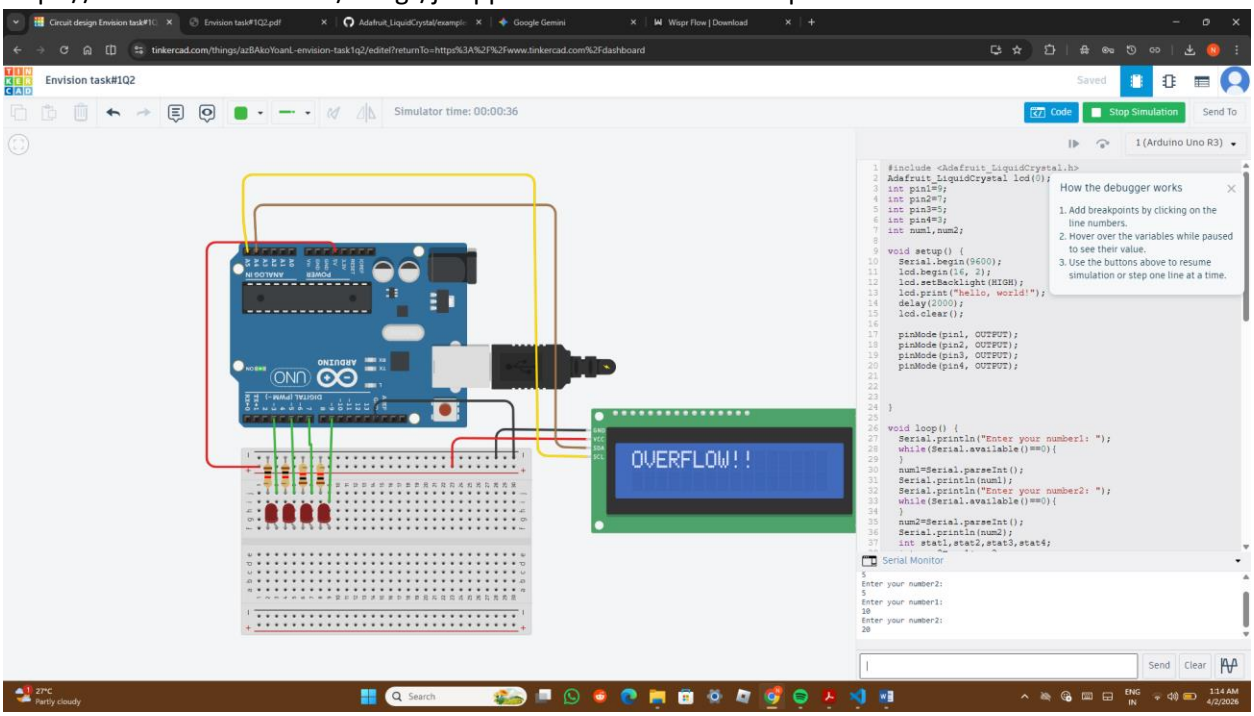
```
while(Serial.available()==0){
}
num2=Serial.parseInt();
Serial.println(num2);
int stat1,stat2,stat3,stat4;
int num3=num1+num2;
if(num3>15){
  lcd.print("OVERFLOW!!");
  delay(4000);
}
else{
  stat4=num3/8;
  int rem1=num3%8;
  stat3=rem1/4;
  int rem2=rem1%4;
  stat2=rem2/2;
  int rem3=rem2%2;
  stat1=rem3;
  if(stat1==1){
    digitalWrite(pin1,HIGH);
  }
  else{
    digitalWrite(pin1,LOW);
  }
  if(stat2==1){
    digitalWrite(pin2,HIGH);
  }
  else{
    digitalWrite(pin2,LOW);
```

```
  }
  if(stat3==1){
    digitalWrite(pin3,HIGH);
  }
  else{
    digitalWrite(pin3,LOW);
  }
  if(stat4==1){
    digitalWrite(pin4,HIGH);
  }
  else{
    digitalWrite(pin4,LOW);
  }
  lcd.print("SUM IS :");
  lcd.setCursor(0,1);
  lcd.print(num3);
  delay(5000);
  lcd.clear();

}
 lcd.clear();
 digitalWrite(pin1,LOW);
 digitalWrite(pin2,LOW);
 digitalWrite(pin3,LOW);
 digitalWrite(pin4,LOW);

}
```

I first initialize the lcd display by setting the backlight to HIGH, and then I display hello world to check if it works(not necessary). And then

initialize the pin modes. In the loop function, I take the input from the user with the help of serial monitor using the serial.parseInt function. The while(serial.available==0) statement makes the serial monitor wait till the user inputs his number t=by pressing enter. I take two inputs, store them in two variabes, and then add them to store in a third variable. Then I check if its more than 15: if yes, then I display overflow message, wait for 4 secs, and then repeat the process…if no, then I divide the num3 by 8, as its an division of integers,the output will be integer(and it can be 0 or 1(0 if number is less than 8, 1 if sum is greater than 8), then store the remainder in another variable, and then repeat the process for 4 and 2. Then I store the results in 4 vribles which will then determine the state of pin(HIGH is 1, LOW if 0)(I used basic if/else for this part of program). Then once the output is displayed, I clear the lcd, and the loop function begins again.





https://www.tinkercad.com/things/jvwpp2wA0c3-envision-task1q1